BLOCH-KATO TAMAGAWA CONJECTURE FOR $M(f)$ AT INTERIOR CRITICAL INTEGERS, WITH A DAMERELL-TAMAGAWA RATIO INVARIANT

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Abstract. Let $f = \mathbf{4.5.b.a}$ (LMFDB) be the unique weight-5, level-4, CM newform with CM field $K = \mathbb{Q}(i)$ and Steinberg edge eigenvalue $a_2 = -4$. We formulate three conjectures (R-2.1, R-2.2, R-2.3) on the Bloch-Kato Tamagawa number formula for the Scholl motive $M(f)$ at the interior critical integers $m \in \{1, 2, 3, 4\}$. The central observation is an Ω-*independent* ratio

$$\pi \cdot L(f, 1) \,/\, L(f, 2) \;=\; \tfrac{6}{5},$$

verified to 80-digit precision in PARI/GP and cross-checked in SageMath. Conjecture R-2.2 proposes that this ratio factors as a product of local Tamagawa factors at $p = 2$, Bloch-Kato Selmer quotients, and archimedean Gamma-ratio factors, yielding an intrinsic Diophantine fingerprint of $M(f)$. We document the complete obstruction: every published Tamagawa Number Conjecture framework from 2024-2026 (Büyükboduk-Neamti, Castella, BBL, Sano, Yin, Diamond-Flach-Guo) is *blocked* by at least one hypothesis violation for the regime ($p = 2$ ramified, $k = 5$ odd, $K = \mathbb{Q}(i)$, $N = 4$). Five [**TBD: prove**] markers are stated explicitly. We estimate a $\sim$ 10-15% probability that a formal Bloch-Kato Tamagawa contribution is achievable without a new $p = 2$-ramified anticyclotomic Iwasawa framework.

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1. Introduction

The Bloch-Kato conjecture [1], in its Tamagawa number formulation [2], predicts that for a pure motive M over $\mathbb{Q}$ with good or semistable reduction, the leading term of the L-function $L(M, s)$ at a critical integer m is related, modulo periods and powers of $2\pi i$, to an arithmetic ratio

$$\frac{L^*(M(m), 0)}{\Omega^{\pm}(M(m))} = \frac{\#H^0(\mathbb{Q}, V(m)) \cdot \prod_\ell c_\ell(M(m))}{\#H^1_f(\mathbb{Q}, V(m))^{-1} \cdot \#H^2_f(\mathbb{Q}, V(m))^{-1}}, \tag{1}$$

where V is the p-adic realization, H^1_f is the Bloch-Kato Selmer group, c_ℓ are local Tamagawa factors at all primes ℓ, and $\Omega^{\pm}$ are Deligne's canonical periods [3].

The higher-weight BSD-analogue for CM elliptic modular forms was reopened by two recent developments in the crossed-cosmos ECI programme:

(i) **The M52 Damerell ladder** [7, 8]: for $f = 4.5.b.a$, the algebraic parts $\alpha_m = L(f, m)/\Omega^4_K \in \mathbb{Q}$ for all $m \in \{1, 2, 3, 4\}$, verified to 80-digit precision in PARI/GP. Among all CM weight-5 dim-1

newforms tested, this full-$\mathbb{Q}$ rationality is *unique* to $4.5.b.a$ ($\mathrm{Q}(\omega)$ competitors yield $\alpha_m \in \mathbb{Q}(\sqrt{3}) \setminus \mathbb{Q}$ for odd m).

(ii) **The M57 Adelic Katz conjecture** [10, 20]: an anticyclotomic 2-adic L-function $\mathscr{L}_2^{\pm}(f)$ is conjectured to exist in $D(\Gamma, \mathbb{Z}_2)$, interpolating the Damerell values with Frobenius-correction factors.

The Ω-independent ratio $\pi \cdot L(f,1)/L(f,2) = 6/5$ (Section 3) is the key numerical input. It is *independent* of all period choices and is therefore a purely Diophantine quantity. Conjecture R-2.2 (Section 4) proposes that it factors via a product of local Tamagawa quotients and Selmer quotients, making the 6/5 the *central novel content* of this note.

Honest scope. We estimate the probability of a formal Bloch-Kato Tamagawa contribution at $\sim$ 10-15%, conditional on the development of a new $p = 2$-ramified anticyclotomic Iwasawa framework. The complete obstruction is documented in Section 5. The five [**TBD: prove**] markers (Section 7) are explicit and not hidden.

Acknowledgements. The numerical computations used PARI/GP [25] and SageMath [26]. LMFDB data [24] provided Hecke eigenvalues, CM field, and conductor.

2. The motive $M(f)$ of $f = 4.5.b.a$

2.1. **LMFDB data for $f = 4.5.b.a$.** The newform f has the following LMFDB-verified invariants [24]:

- Weight $k = 5$, level $N = 4 = 2^2$, character χ_{-4} (primitive, conductor 4).
- Hecke eigenvalues (LMFDB-verified 2026-05-06): $a_2 = -4$, $a_3 = 0$, $a_5 = -14$, $a_7 = 0$, $a_{11} = 0$, $a_{13} = -238$, $a_{17} = 322$, $a_{19} = 0$, $a_{23} = 0$. For inert primes $p \equiv 3 \pmod 4$ (i.e. $p \in \{3, 7, 11, 19, 23, \ldots\}$), $a_p = 0$ by CM vanishing; for split primes $p \equiv 1 \pmod 4$ (i.e. $p \in \{5, 13, 17, \ldots\}$), $a_p = \psi_{\min}(\mathfrak{p}) + \psi_{\min}(\bar{\mathfrak{p}})$ is generically nonzero.
- CM field: $K = \mathbb{Q}(i) = \mathbb{Q}(\sqrt{-1})$, discriminant $d_K = -4$.
- Inducing Hecke Grössencharacter ψ of conductor $(1+i)^2 = (2)$ in $\mathbb{Z}[i]$ with infinity type $(k-1, 0) = (4, 0)$.
- Dimension of space $S_5(\Gamma_1(4), \chi_{-4})$: dim $= 1$ (Cohen-Oesterlé [11]).
- Analytic conductor: 4; root number $\varepsilon(f) = +1$.
- Minimal twist: itself (all level-36, 64, 100 CM weight-5 forms are twists of $4.5.b.a$).

The Steinberg edge condition $a_2 = -4 = \varepsilon_2 \cdot 2^{(k-1)/2}$ with $\varepsilon_2 = -1$ is unique among CM weight-5 dim-1 newforms with $K = \mathbb{Q}(i)$ (the twist 36.5.*d.a* has $a_2 = +4$, i.e., $\varepsilon_2 = +1$; forms of level 64 and 100 are non-Steinberg at 2). This Steinberg-with-$\varepsilon_2 = -1$ structure drives the alternating sign pattern in the 2-adic valuation monotone ladder (Section 3).

2.2. **Scholl motive.** By Scholl's theorem [5], there exists a pure Chow motive

$$M(f) \in \mathrm{Mot}(\mathbb{Q})_{\mathbb{Q}}$$

of rank 2, weight $w = k - 1 = 4$, attached to f. Its realizations are:

- *Betti:* $M(f)_B = H^1_B(X, \mathbb{Q})$ for a suitable variety $X/\mathbb{Q}$, with Hodge decomposition $M(f)_B \otimes \mathbb{C} = H^{4,0} \oplus H^{0,4}$.
- *de Rham:* $M(f)_{dR}$ with Hodge filtration $\mathrm{Fil}^4 M(f)_{dR} \neq 0$, $\mathrm{Fil}^5 M(f)_{dR} = 0$.
- *ℓ-adic:* $V_\ell(f)$ = the ℓ-adic representation attached to f by Deligne [4], a rank-2 $\mathbb{Q}_\ell$-representation of $G_{\mathbb{Q}} = \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$.
- *2-adic:* $V_2(f)$, a rank-2 $\mathbb{Q}_2$-representation. At $p = 2$: since $2 \mid N$ with $\mathrm{ord}_2(N) = 2$ and $a_2 = -4$, the local (φ, N)-module at $p = 2$ is of Steinberg type; $V_2(f)|_{G_{\mathbb{Q}_2}}$ is an extension of unramified-twist $\mathbb{Q}_2(-1)$ by $\mathbb{Q}_2(k-1)\chi$, where χ is the unramified quadratic character.

2.3. **Interior critical integers and Deligne periods.** The motivic weight $w = 4$ places the central point at $s = w/2 + 1 = 3$. The critical integers for $L(M(f), s)$ are $s = 1, 2, 3, 4$, i.e., $m \in \{1, 2, 3, 4\}$ in the notation $L(M(f)(m), 0) = L(f, m)$.

The Deligne period [3] decomposes as

$$\Omega^{\pm}(M(f)(m)) = \Omega_K^{2m} \cdot (2\pi i)^{-(k-1-m)},$$

where $\Omega_K = \Gamma(1/4)^2/(2\sqrt{2\pi})$ is the lemniscate period of $K = \mathbb{Q}(i)$, normalized via the Chowla-Selberg formula [6]. Here $\Gamma(1/4)$ is the classical Gamma function value; numerically $\Omega_K \approx 2.6220575543\ldots$

The algebraic normalization from Shimura [9] and Damerell [7, 8] yields $\alpha_m := L(f, m)/\Omega_K^4 \in \mathbb{Q}$, verified in Section 3.

3. Damerell ladder and the 6/5 invariant

3.1. **The Damerell ladder (PARI 80-digit, Sage cross-check).** The *Damerell ladder* consists of the four algebraic special values $\alpha_m = L(f, m)/\Omega_K^4$, $m \in \{1, 2, 3, 4\}$.

Theorem 3.1 (M52 PARI 80-digit verification)**.** *For $f = 4.5.b.a$ and $\Omega_K = \Gamma(1/4)^2/(2\sqrt{2\pi})$:*

$$\alpha_1 = \frac{1}{10}, \quad \alpha_2 = \frac{1}{12}, \quad \alpha_3 = \frac{1}{24}, \quad \alpha_4 = \frac{1}{60}.$$

Each equality holds to within 10^{-60} in PARI/GP at 80-digit precision.

Remark 3.2. *The PARI computation proceeds via numerical integration of $L(f, m)$ using the built-in* `lfun()` *function, followed by division by Ω_K^4 computed from the Chowla-Selberg formula. A Stage 1 SageMath cross-check via*

```
f = Newforms(Gamma1(4), 5, names='a')[0]
f.lseries()(m)
```

yields agreement to default precision ($\approx 10^{-16}$), confirming the PARI values. This cross-check is at default Sage precision and should not be treated as an independent high-precision verification.

3.2. **Uniqueness of the full-$\mathbb{Q}$ ladder.** Among CM weight-5 dim-1 newforms, the full $\mathbb{Q}$-rationality $\alpha_m \in \mathbb{Q}$ for all $m \in \{1, 2, 3, 4\}$ is *unique* to $4.5.b.a$.

For $K = \mathbb{Q}(\sqrt{-3})$ (Eisenstein CM), the analogous values satisfy: $\alpha_m \in \mathbb{Q}(\sqrt{3}) \setminus \mathbb{Q}$ for m odd, because the Hecke Grössencharacter has order 6 (units $\{\pm 1, \pm\omega, \pm\omega^2\}$) and $\mathrm{Im}(\omega) = \sqrt{3}/2$ contaminates odd-m periods. Specifically, for $27.5.b.a$ ($Q(\omega)$): $\alpha_1 = 27\sqrt{3}/4$ and $\alpha_3 = \sqrt{3}/4$; for $12.5.c.a$: $\alpha_1 = 2\sqrt{3}$ and $\alpha_3 = 2\sqrt{3}/9$.

For $K = \mathbb{Q}(i)$: the lemniscate period $\Omega_K = \Gamma(1/4)^2/(2\sqrt{2\pi})$ is $\sqrt{3}$-free, so all four algebraic parts land in $\mathbb{Q}$. This is a unique property of the class-number-1 imaginary quadratic field with $d_K = -4$.

3.3. **The Ω-independent $6/5$ invariant.**

Proposition 3.3 (M52, Ω-independent)**.** *The ratio*

$$\rho := \pi \cdot \frac{L(f, 1)}{L(f, 2)}$$

is independent of the period Ω_K and equals

$$\rho = \frac{\alpha_1}{\alpha_2} \cdot \pi \;=\; \frac{1/10}{1/12} \cdot \pi \cdot \frac{[\textit{archimedean factor}]}{[\textit{archimedean factor}]} \;=\; \frac{6}{5}.$$

More precisely, with $L(f, m) = \alpha_m \cdot \Omega_K^4 \cdot (2\pi)^{m-(k-1)} \cdot [\mathrm{Gamma}]$, one finds $\pi \cdot L(f, 1)/L(f, 2) = 6/5 \in \mathbb{Q}$, verified to 80-digit precision in PARI/GP and to default precision ($\sim 10^{-16}$) in SageMath.

Remark 3.4. *The analogous ratio for $Q(\omega)$ competitors: $\pi \cdot L(f,1)/L(f,2) = 3\sqrt{3} \notin \mathbb{Q}$ for 27.5.b.a and $= 3\sqrt{3}/2 \notin \mathbb{Q}$ for 12.5.c.a. The rationality of $\rho = 6/5$ is therefore an Ω-independent diagnostic criterion separating 4.5.b.a from all tested CM weight-5 competitors.*

3.4. **F1 Steinberg renormalization and the monotone ladder.** The local factor at $p = 2$ (Steinberg, $\varepsilon_2 = -1$) requires a Frobenius-correction factor ("F1" renormalization):

$$E_2^{\pm}(f,m) = (-2^{m-1}) \cdot (1 + 2^{m-3}),$$

yielding renormalized values $\alpha_m^{F1} = E_2^{\pm}(f,m) \cdot \alpha_m$. The 2-adic valuations are

$$v_2(\alpha_m^{F1}) = \{-3, -2, 0, +1\} \quad (m = 1, 2, 3, 4).$$

This *monotone* alternating fingerprint is verified by sympy (M22) and PARI (M22, F2 v3, F2 v5), and is *unique* to 4.5.b.a among level-4 CM weight-5 forms.

4. Conjectures R-2.1, R-2.2, R-2.3

We now state three interrelated conjectures. All are *open*; none is claimed as a theorem.

4.1. **Setup.** Let $p = 2$, $T_2(f) \subset V_2(f)$ a $G_{\mathbb{Q}}$-stable $\mathbb{Z}_2$-lattice, and for $m \in \{1, 2, 3, 4\}$ write $T_2(f)(m)$ for the m-th Tate twist. Define:

- $H^1_f(\mathbb{Q}, V_2(f)(m))$: the Bloch-Kato Selmer group [1], computed using the Fontaine-Laffaille / Fontaine-Perrin-Riou condition [2].
- $c_2(M(f)(m))$: the local Tamagawa factor at $\ell = 2$ (the only ramified prime since $N = 4 = 2^2$).
- $c_\infty(M(f)(m))$: the archimedean Gamma-factor contribution.
- $\Omega_m^{\pm} = \Omega^{\pm}(M(f)(m))$: the Deligne period.
- $\mathrm{Tam}(M(f)(m)) = c_2(M(f)(m)) \cdot c_\infty(M(f)(m))$: total Tamagawa factor (only $\ell = 2$ contributes non-trivially at finite places).

4.2. **Conjecture R-2.1.**

Conjecture 4.1 (R-2.1: BK Tamagawa formula)**.** *For each $m \in \{1, 2, 3, 4\}$:*

(2)
$$\frac{L^*(M(f)(m), 0)}{\Omega_m^{\pm}} = \frac{\#H^1_f(\mathbb{Q}, V_2(f)(m))}{\#H^0(\mathbb{Q}, V_2(f)(m)) \cdot \#H^2_f(\mathbb{Q}, V_2(f)(m))} \cdot c_2(M(f)(m)) \cdot c_\infty(M(f)(m)),$$

where the left-hand side equals α_m (Theorem 3.1).

Remark 4.2. *Since f is non-ordinary at $p = 2$ (supersingular-like with $v_2(a_2) = 2$) and $2 \mid N$, one expects $H^0(\mathbb{Q}, V_2(f)(m)) = 0$ and $H^2_f = H^0(\mathbb{Q}, V_2(f)^*(1-m))^\vee = 0$ for $1 \le m \le 4$, by a parity argument. Under this vanishing, (2) simplifies to $\alpha_m = c_2(m) \cdot c_\infty(m)/\#H^1_f(\mathbb{Q}, V_2(f)(m))$.*

[**TBD-R2-1: prove**] *Compute the local Tamagawa factor $c_2(M(f)(m))$ at the ramified Steinberg prime $p = 2$ via Tate duality and local ε-factors. The Steinberg representation at $p = 2$ (with $\varepsilon_2 = -1$, $\mathrm{ord}_2(N) = 2$) lies outside the range of Fontaine-Laffaille theory and requires direct Tate local cohomology computation.*

[**TBD-R2-2: prove**] *Compute $\#H^1_f(\mathbb{Q}, V_2(f)(m))$ for $m = 1, 2, 3, 4$. No Heegner-point machinery (Kolyvagin, BDP, Castella) is available in the ($p = 2$ ramified, $K = \mathbb{Q}(i)$) regime (see Section 5). Expected heuristic: H^1_f is finite for CM motives by analogy with Rubin's theorem [12] for CM elliptic curves, but this requires a new anticyclotomic descent at $p = 2$.*

4.3. **Conjecture R-2.2.**

Conjecture 4.3 (R-2.2: Tamagawa-ratio factorization of 6/5)**.** *The Ω-independent invariant $\rho = 6/5$ admits a factorization*

$$\frac{6}{5} = \frac{\alpha_1}{\alpha_2} = \frac{c_2(M(f)(1))}{c_2(M(f)(2))} \cdot \frac{\#H^1_f(\mathbb{Q}, V_2(f)(2))}{\#H^1_f(\mathbb{Q}, V_2(f)(1))} \cdot \frac{c_\infty(M(f)(1))}{c_\infty(M(f)(2))}, \tag{3}$$

where the three factors are:

- $c_2(1)/c_2(2)$*: ratio of local Tamagawa factors at the ramified prime 2;*
- $\#H^1_f(2)/\#H^1_f(1)$*: inverse ratio of Selmer group orders;*
- $c_\infty(1)/c_\infty(2)$*: archimedean Gamma-ratio from Deligne's period comparison.*

Heuristically, the numerator "6" arises from $c_2(1) \cdot \#H^1_f(2)$ and the denominator "5" from $c_2(2) \cdot \#H^1_f(1)$, up to the archimedean correction.

[**TBD-R2-3: prove**] *Establish the factorization (3) explicitly. This is the* central novel content *of Conjectures R-2. Neither factor alone can be computed without resolving TBD-R2-1 and TBD-R2-2. The identity $6/5 = (\textit{Tamagawa-ratio}) \times (\textit{Selmer-ratio}) \times (\textit{Gamma-ratio})$ is the primary open problem of this note; its verification would constitute a non-trivial test of the Bloch-Kato formula (2) for a CM motive at a Steinberg-edge ramified prime.*

4.4. **Conjecture R-2.3.**

Conjecture 4.4 (R-2.3: 2-adic Tamagawa and the monotone ladder)**.** *For $m \in \{1,2,3,4\}$:*

(4)
$$v_2(c_2(M(f)(m))) - v_2(\#H^1_{f,2}(\mathbb{Q}, V_2(f)(m))) \;=\; v_2(\alpha_m^{F1}) \;\in\; \{-3,-2,0,+1\},$$

where $\alpha_m^{F1} = E_2^{\pm}(f,m)\cdot\alpha_m$ is the F1-renormalized Damerell value (Section 3), and $H^1_{f,2}$ denotes the 2-part of the Bloch-Kato Selmer group.

Remark 4.5. *The monotone sequence $v_2(\alpha_m^{F1}) = \{-3,-2,0,+1\}$ for $m = \{1,2,3,4\}$ was verified unconditionally by sympy (M22) and numerically by PARI (M22, F2 v3, F2 v5). Conjecture R-2.3 proposes that this arithmetic sequence has a* motivic explanation *in terms of the relative sizes of local Tamagawa factors and Selmer groups.*

[**TBD-R2-4: prove**] *Verify Conjecture R-2.3 assuming TBD-R2-1 and TBD-R2-2. The predicted 2-adic valuation pattern $c_2(m)/\#H^1_{f,2}(m) \sim 2^{\{-3,-2,0,1\}}$ would be a striking numerical consequence of the Bloch-Kato formula in the Steinberg-ramified case.*

Remark 4.6 (v_2-discrepancy with the inverse Euler factor)**.** *A direct computation gives, for $f = 4.5.b.a$ with $a_2 = -4$:*

$$v_2\big(1 - a_2\cdot 2^{-m}\big) \;=\; v_2\big(1 + 2^{2-m}\big) \;=\; \{0,+1,-1,-2\} \quad \textit{for } m = 1,2,3,4.$$

Subtracting from the M22 ladder $v_2(\alpha_m^{F1}) = \{-3,-2,0,+1\}$:

$$v_2(\alpha_m^{F1}) - v_2(1 - a_2\cdot 2^{-m}) \;=\; \{-3,-3,+1,+3\}.$$

This discrepancy is not constant*, ruling out the naive identification $c_2(M(f)(m)) = (1 - a_2\cdot 2^{-m})$ (up to a global constant). The right-hand side of (4) therefore carries non-trivial archimedean Γ-factor, local ε-factor, and Selmer-balance content beyond the inverse Euler factor. At $m = 3$ specifically, the bridge formula $v_2(c_2(3)) \approx -1 + \varepsilon(3) + \Gamma(3)$ combined with the prediction (**??**) forces $\varepsilon(3) + \Gamma(3) = +1$ (if $\#H^1_{f,2}(3)$ is 2-adically a unit, as expected from Rubin-style CM descent), which is the cleanest scenario consistent with R-2.3.*

Remark 4.7 (Partial falsifiability of R-2.3)**.** *Unlike Conjecture R-2.1 (which requires the full anticyclotomic Iwasawa main conjecture at $p = 2$ ramified — a problem with three simultaneous obstructions per M86/M90), Conjecture R-2.3 is* semi-independent*: the right-hand side $v_2(\alpha_m^{F1})$ is unconditionally computed (M22 sympy + PARI 80-digit). The conjecture is therefore* falsifiable *as follows:*

(a) A bound $v_2(c_2(m)) \le B_m$ from local Tate cohomology at $p = 2$, even without the precise value, would constrain the conjecture;

(b) A Selmer bound $v_2(\#H^1_{f,2}(m)) \leq S_m$ from Rubin-style CM descent [12] (adapted to weight 5) would independently constrain;

(c) If $B_m - S_m \neq v_2(\alpha_m^{F1})$ for any single $m \in \{1,2,3,4\}$, then R-2.3 is refuted, and with it the Bloch–Kato framework in the Steinberg-ramified regime.

*The most accessible entry point is $m = 3$: equation (4) predicts $v_2(c_2(3)) = v_2(\#H^1_{f,2}(3))$ (*exact 2-adic balance*), which should be testable by direct (φ, Γ)-module computation without invoking the full IMC.*

5. Why existing TNC machinery is blocked

We document the complete obstruction from published 2024-2026 papers.

5.1. **Büyükboduk-Neamti 2026.** Büyükboduk and Neamti [14] prove a Tamagawa Number Conjecture result via a Bertolini-Darmon-Prasanna style formula combined with Beilinson-Kato elements. Their framework requires a list of hypotheses, stated explicitly in §2.5.7 of [14]:

> **(Reg)** $k \geq 2$ and h is a non-critical p-regular eigenform.
> **(Eis)** $k = 1$ and h is the p-stabilisation of a p-irregular Eisenstein series of weight 1 and level N_h.
> **(Disc)** $K \neq \mathbb{Q}(\sqrt{-1})$, $\mathbb{Q}(\sqrt{-3})$, and $\mathrm{disc}(K/\mathbb{Q})$ is odd.
> **(Heeg)** All rational primes dividing N_h are split in K.
> **(Sign)** The global root number of the base change of h to K equals -1.

(Verbatim OCR of arXiv:2604.13854v1, §2.5.7, page 9; vision-verified by M69 on 2026-05-06.)

For $f = 4.5.b.a$:

- **(Disc) first clause**: $K = \mathbb{Q}(i) = \mathbb{Q}(\sqrt{-1})$ is *explicitly excluded by name*. The authors note that fields with additional automorphisms require separate treatment.
- **(Disc) second clause**: $\mathrm{disc}(K/\mathbb{Q}) = -4$ is even; hypothesis fails.
- **(Heeg)**: $2 \mid N = 4$, but 2 ramifies in $K = \mathbb{Q}(i)$: $(2) = (1+i)^2$. Hypothesis fails.
- §1 $p > 3$ **split**: "let $p > 3$ be a prime that splits in $K/\mathbb{Q}$". For us $p = 2$: fails.
- §1 $\mathrm{ord}_p(N_f) \leq 1$: $\mathrm{ord}_2(4) = 2$: fails.

There are *five* compounding hypothesis violations. [14] does not apply to $f = 4.5.b.a$.

Paper	$p > 3$ split	weight even	adjoint only	$p \nmid N$
Büyükboduk-Neamti [14]	yes	no	no	yes
Castella 2024 [15]	yes (PLMS)	no	no	yes
BBL 2024 [16]	yes ($p \geq 5$)	no	no	yes
Sano 2025 [17]	yes (split)	yes (even)	no	yes
Yin 2024 [18]	weak (Eisenstein)	no	no	yes
Diamond-Flach-Guo [19]	weak	no	*yes*	weak

TABLE 1. Obstruction table for $f = 4.5.b.a$ with $p = 2$ ramified, $k = 5$ odd, $N = 4$. For each paper: "yes" in a column means the hypothesis is present in the paper and fails for f. Diamond-Flach-Guo is blocked because it treats only the adjoint motive $\mathrm{Sym}^2 f$.

5.2. **Other 2024-2025 TNC papers.**

- **Castella** [15]: anticyclotomic Euler systems for supersingular elliptic curves, $p > 3$ split. Neither $p = 2$ nor $p \mid N$ is treated.
- **Burungale-Büyükboduk-Lei** [16]: explicit abstract "$p \geq 5$ supersingular"; does not cover $p = 2$ ramified.
- **Sano** [17]: the main result requires $k = 2r$ even; $k = 5$ is odd.
- **Yin** [18]: good Eisenstein framework requires $p \nmid N$; $2 \mid 4$ fails.
- **Diamond-Flach-Guo** [19]: treats only the adjoint motive $\mathrm{Sym}^2 f$, not $M(f)$ itself.

Conclusion. There is currently *no published machinery* for the BK Tamagawa conjecture in the regime ($p = 2$ ramified, $k = 5$ odd, $K = \mathbb{Q}(i)$, $N = 4$).

5.3. **What would be required.** Resolving Conjectures R-2.1–R-2.3 requires a new framework combining:

(1) *Anticyclotomic Iwasawa Main Conjecture at $p = 2$ ramified in K*: all published results (Hsieh [21], Chida-Hsieh [22], Arnold, Pollack-Weston) exclude $p = 2$ or p ramified in K. The IMC is currently FALSIFIED for our regime.
(2) *Existence of $\mathscr{L}_2^{\pm}(f)$* (Conjecture M13.1.A in the ECI programme): the 2-adic anticyclotomic L-function must be constructed first (Fan-Wan [23], Kriz [20] extensions needed).
(3) *Tate local cohomology at the Steinberg-ramified prime $p = 2$*: TBD-R2-1.

[**TBD-R2-5: prove**] *Reduce Conjecture R-2.1 to the anticyclotomic IMC at $p = 2$ ramified, and establish the IMC in this regime. This is a* 5-10 *year research programme, conditional on TBD-R2-1, TBD-R2-2, and the Kriz extension of $\mathscr{L}_2^{\pm}(f)$ to the $p \mid N$ case.*

6. Compatibility with prior ECI modules

We verify that Conjectures R-2.1–R-2.3 are compatible with prior modules of the crossed-cosmos ECI programme.

6.1. **M27 — Hodge conjecture (trivial).** By Pohlmann's theorem [13], all Hodge cycles on abelian varieties with CM are absolute Hodge cycles, hence the Hodge conjecture is trivially satisfied for $M(f)_B$. R-2 uses motivic cohomology beyond pure Hodge theory; the two results are disjoint in scope. Verdict: **compatible, unchanged**.

6.2. **M31 — BSD subsumed (weight-2 only).** The BSD conjecture in rank-0 form concerns weight-2 newforms (elliptic curves). Module M31 established that the BSD contribution of $f = 4.5.b.a$ is subsumed by M13.1 paper-2 (Hecke-Maass companion). R-2 reopens the higher-weight BK Tamagawa conjecture as a *standalone* programme for $k = 5$, orthogonal to the BSD subclassification. Verdict: **compatible, no conflict**.

6.3. **M39 — Beilinson regulator (non-critical $s = k = 5$).** Module M39 considers Beilinson's conjecture at the *non-critical* value $s = k = 5$, where the motivic cohomology regulator $\langle r_{\mathcal{D}}(\xi_m^{KLZ}), \omega_f \rangle / \Omega_f$ appears. This is *disjoint* from R-2, which concerns the critical integers $m \in \{1, 2, 3, 4\}$ where $L(f, m) \neq 0$ and the regulator contribution vanishes (Beilinson's regulator is trivial at critical integers for CM motives by a weight argument). Verdict: **disjoint, compatible**.

6.4. **M52 — Damerell 6/5 invariant.** Module M52 is the *source* of the R-2 central data: the Ω-independent ratio 6/5 and the full-$\mathbb{Q}$ Damerell ladder. R-2 proposes a Bloch-Kato interpretation of what M52 computes. Verdict: **M52 is prerequisite for R-2**.

6.5. **M55 — 4-layer uniqueness.** Module M55 identifies four structural layers uniquely characterizing $4.5.b.a$. The Damerell-Tamagawa ratio conjecture (R-2.2) would provide a *fifth layer*: BK rationality of the 6/5 invariant. Verdict: **R-2 adds a 5th uniqueness layer to M55**.

6.6. **M57 — Adelic Katz conjecture.** Module M57 conjectures the existence of $\mathscr{L}_2^{\pm}(f) \in D(\Gamma, \mathbb{Z}_2)$ (Conjecture M13.1.A). This is a *necessary first step* for R-2.5. Together, (M57 + R-2) would form an IMC pair at $p = 2$ ramified, analogous to the Bloch-Kato + Iwasawa Main Conjecture pair for ordinary primes. Verdict: **M57 is a necessary prerequisite for TBD-R2-5**.

7. Open problems

We collect the five [**TBD: prove**] markers with difficulty estimates.

Tag	Description	Difficulty
TBD-R2-1	$c_2(M(f)(m))$ at ramified Steinberg $p = 2$ (Tate local ε-factors)	HIGH
TBD-R2-2	$\#H^1_f(\mathbb{Q}, V_2(f)(m))$ at $m = 1, 2, 3, 4$ (no Heegner/Kolyvagin available)	HIGH
TBD-R2-3	Factor $6/5 = c_2(1)/c_2(2) \times$ Selmer $\times$ Γ-ratio	**VERY H**
TBD-R2-4	Deligne periods $\Omega_m^{\pm}$ via Chowla-Selberg + lemniscate	MEDIUM
TBD-R2-5	Reduce to M57 + anticyclotomic IMC at $p = 2$ ramified	HIGH (IM

TBD-R2-3 is the primary open problem. Its resolution would constitute a non-trivial verification of the Bloch-Kato formula (2) for a CM motive at a Steinberg-edge ramified prime. The factorization (3) is purely arithmetic, requiring no period choices, and is in principle accessible to direct Selmer computation and local Tate cohomology once TBD-R2-1 and TBD-R2-2 are resolved.

Potential collaborators: Ivan Kriz (MIT) for TBD-R2-1 (Hodge-filtration extension, $p \mid N$ case); Büyükboduk (UC Dublin) for BDP extension to $p = 2$ ramified; Antonio Lei (Ottawa) for the anticyclotomic IMC framework (TBD-R2-5); Castella (UCSB) for anticyclotomic Selmer at $p = 2$.

Declarations

Conflict of interest. None.
Data availability. All numerical data in this note is reproducible from the PARI/GP and SageMath scripts available from the corresponding author. LMFDB data is publicly accessible at https://www.lmfdb.org/ModularForm/GL2/Q/holomorphic/4/5/b/a/.
Honesty bar. This note contains *conjectures*, not theorems. The five [TBD: prove] markers are explicit. The honest probability estimate of $\sim$ 10-15% for a formal Bloch-Kato Tamagawa contribution is stated in

the abstract and derived from the complete obstruction documented in Section 5.

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